



Vivek Mishra <vivekmishraima@gmail.com>

Invitation to review submission for The 3rd International Conference on Artificial Intelligence, Computer, Data Sciences and Applications

Microsoft CMT <noreply@msr-cmt.org>
To: Vivek Mishra <vivekmishraima@gmail.com>

Fri, Nov 7, 2025 at 9:32 PM

Dear Colleague,

We hope this email finds you well. We are writing to kindly request your comments on a paper that was recently submitted to the 3rd International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA) which will take place in Boracay Island, Philippines on 05-07 February 2026.

www.acdsa.org

We would greatly appreciate it if you could provide your comments on the paper at your earliest convenience. Please address aspects such as originality, methodology, results, organization, clarity, technical soundness, and any strengths or weaknesses you identify. Please try to write at least 20 words for your evaluation.

Please send your comments no later than 12 November 2025.
Thank you for your cooperation, support and kind help.

Title:

1105: A Multi-Scenario Machine Learning Framework for Predicting 5G Network Quality of Service

Abstract:

Fifth-generation (5G) mobile networks demand dynamic and predictive Quality of Service (QoS) management mechanisms to meet high data rate, low latency, and reliability requirements. However, existing network monitoring systems are predominantly reactive, resulting in inefficiencies in resource allocation and latency optimization. This study proposes an advanced machine learning-based modeling framework for estimating key QoS parameters in 5G networks. The framework models four critical QoS indicators—allocated bandwidth, bandwidth difference, end-to-end latency, and resource allocation ratio—using a dataset of 399 samples collected from real network recordings. Data preprocessing included unit normalization, interaction-based feature extraction, and application-level priority mapping. Five regression algorithms (Random Forest, Gradient Boosting, XGBoost, Ridge, and Linear Regression) were compared. The best-performing models achieved the following results: bandwidth allocation with XGBoost ($R^2 = 0.99$), bandwidth difference with Gradient Boosting ($R^2 = 0.7913$), delay estimation with Random Forest ($R^2 = 0.99$), and resource allocation with Linear Regression ($R^2 = 1.00$). The experimental results demonstrate that the proposed framework achieves high prediction accuracy in critical tasks such as proactive resource scheduling, delay management, and bandwidth optimization. This approach provides data-driven, predictive decision support for network operators, enhancing QoS reliability and significantly improving overall network efficiency.

Click the link below to accept.

<https://cmt3.research.microsoft.com/ACDSA2026/User/AcceptInvitationForSubmission?uniqueId=349247d3-22ea-473a-89e3-f62d539e2e49>.

Click the link below to decline.

<https://cmt3.research.microsoft.com/ACDSA2026/User/DeclineInvitationForSubmission?uniqueId=349247d3-22ea-473a-89e3-f62d539e2e49>.

Here is the paper template: http://www.acdsa.org/ACDSA_template.docx

*** PLEASE NOTE: This is a two-step process. After clicking on the link, you will be directed to a webpage to confirm. ***

*** If you are not able to see the paper, please inform us so that we can send you the submission files ***

The conference chair will send you a reviewer certificate for your valuable contribution to ACDSA'25.

11/9/25, 12:50 AM

Gmail - Invitation to review submission for The 3rd International Conference on Artificial Intelligence, Computer, Data Sciences and...

We eagerly await your comments and feedback.
Best regards,

ACDSA'26 Organizing Committee
www.acdsa.org

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our [Privacy Statement](#).

Microsoft Corporation
One [Microsoft Way](#)
[Redmond, WA 98052](#)